CPLREADER - A CFD DATA READER AND ACOUSTIC SOURCE TERM GENERATOR FOR CFS++

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Contents

1 Name

cplreader - A CFD Data Reader and Acoustic Source Term Generator for *CFS++*.

The name was invented by Max Escobar , the original author of this program, during his PhD thesis [8].

2 Usage

3 Description

In computational aero-acoustics the so-called hybrid methods are quite common. These methods try to extract source terms for acoustics from CFD data. There exits a wide variety of such methods. The volume based approach used by *CFS++*[1] [13] is the computation of Lighthill source terms in a FEM framework directly on the the CFD mesh. Since the CFD mesh is usually too fine to perform an acoustic computation on it, the source terms have to be conservatively interpolated to a dedicated acoustic grid. The first step is the job of *cplreader*. The interpolation is performed within *CFS++*.

The source term generator *cplreader* is used to read CFD meshes from codes like *ANSYS CFX*[2], *OpenFOAM*[3] or *FASTEST-3D*[4] with the associated velocity, pressure and density fields, to compute the source terms and writing them to *HDF5*[5] files.

Aside from its primary designated use *cplreader* can also be used to convert meshes from the *ANSYS* mesh generator to the *HDF5* file format for use in *CFS++* or *NACS*.

[1] *CFS++* is a FEM solver developed for research purposes at LSE Erlangen and the Chair of Applied Mechatronics at Klagenfurt. There exists also a commercial version called *NACS* which is distributed by Simetris GmbH http://www.simetris.de

[2] ANSYS CFX, http://www.ansys.com/Products/cfx.asp

[3] OpenFOAM, http://www.opencfd.co.uk/openfoam

[4] Flow solver FASTEST-3D developed at LSTM Erlangen

[5] Hierarchical Data Format 5, http://hdf.ncsa.uiuc.edu/HDF5

4 Options

Table 1: Command line options of *cplreader*.

Parameter	Default Value	Short Description
--docu		Create directory with detailed documentation (can just be PDF) This option will create the directory `cplreader_docs` in the current working directory and put the file `cplreader_docu.pdf` there.
--name		Name of dataset as well as of the directory containing the dataset This is the name of the dataset to be worked on. For most readers the data files must be placed in a directory of this name, too.
--type	CFX	Type of dataset (can be ANSYS | CFS++ | CFX | CFX_EXPORT | FASTEST | OPENFOAM | CGNS | ENSIGHT | FIELDVIEW) Specify the type of dataset to be read: ANSYS see Sec. 8.1, CFX see Sec. 7.2, CFX_EXPORT see Sec. 7.3, FASTEST see Sec. 7.1, OPENFOAM see Sec. 7.4.
--xmlfile		XML file containing options for cplreader. The XML file specified here contains informations for the file readers/writers.

--basedir	.	Base directory where subdirectories are searched.
		The files to be read are searched for in the directory specified by basedir.
		For CFX this means for example:
		
		`basedir/name/`
		`|-- name`
		`| |-- 1.trn`
		`| +-- 2.trn`
		`|-- name.def`
		`+-- name.res`
		
		By default the files will be searched for in the current working directory.

--extfiles	true	Use external time step files (just like CFX .trn files). If this parameter is true *cplreader* will create on master .h5 file containing the mesh and just references to the time step files. For each time step an extra *external* .h5 file will be created. If this parameter is false everything goes into one big .h5 file. It may be tricky to copy such big files to USB hard disks, which are usually formatted with a FAT32 file system.
--dim	3	Dimension of fluid data. (can be 2 | 3) This parameter defines the geometrical dimension of the fluid data set. Most readers do not require this parameter. It is however present for the FASTEST reader.
--numsteps	0	Number N of timesteps to read. Default: read all available steps. The number of timesteps to read. By default (0) all available timesteps will be read.
--activeparts	all	Values will only be output on partitions specified by activeparts (all | numbers seperated by ␣ or ; or |). When dealing with very large data sets with many partitions/sub-domains, it is often convenient to read the fields and compute the acoustic source for just a few of them.
--outputfields	acouRhsLoad	Which physical fields should be output ([acouDivLighthillTensor | acouRhsLoad | acouRhsLoadDensity | fluidMechDensity | fluidMechPressure | fluidMechVelocity | fluidMechTKE]). Values may be separated by ␣ or ; or | Sometimes it may be necessary to do some post-processing on the fluid fields to get some understanding of the problem at hand. For this reason it is possible to write the most important fields (velocity, pressure, source terms, turbulence kinetic energy, etc.) to the .h5 files by using the outputfields parameter.
--firststep	1	Index of first time step (only for CFX_EXPORT, FASTEST) To start reading at a different time step than the first one this parameter can by used. It is just implemented for CFX_EXPORT and FASTEST at the moment.
--stepincr	1	Step increment for reading the files Increment time step counter by this value und just read each stepincr-th time step.
--timestep	-1.0	Time step length T in seconds This parameter specifies the length of each timestep in seconds.
--calcsrc	false	Calculate the acoustic sources from velocity If this parameter is set a velocity field has to present in each time step from which the Lighthill source term will be computed.
--geomscale	1.0 1.0 1.0	Scale factors for geometry in the format factorX factorY factorZ

Continued on next page

Table 1 – continued from previous page

Parameter	Default Value	Short Description
		This parameter specifies the factors to be multiplied with the corresponding components of the nodal coordinates. This can be useful, if the CFD computation used a different scaling (cf. OpenFOAM simulations of the Freiberg guys)
--velscale	1.0 1.0 1.0	Scale factors for velocity field in the format factorX factorY factorZ This parameter specifies the factors to be multiplied with the corresponding components of the velocity field. This can be useful, if the CFD computation used a different scaling (cf. OpenFOAM simulations of the Freiberg guys)
--density	1.0	Density of the fluid. Default density is 1.0 for testing. The mean density of the fluid may be specified by this parameter. Some practically relevant values include: air $1.2041kg/m^3$ (at $20°C$ and $101.325kPa$), water $998.2kg/m^3$ (at $20°C$)
--segfault	true	Cause program to segfault if an exception is encountered. This parameter is especially important for debugging purposes and should not be necessary in standard runs.
--deltemp	true	Delete temporary files. Some file readers (ANSYS) create or use temporary files. This parameter determines, whether those files will be deleted or not.
--outprec	double	Write data as single or as double precision (can be single | double) With outprec the user can specify the precision with which the floating point numbers will be written to the .h5 files. specifying *single* here may save some space but the user has to be sure what he is doing!
--justinit	false	Just perform initialization step. Specifying justinit will let *cplreader* just perform the init phase. This comes in handy when developing a new reader.
--justmesh	false	Just convert the mesh. Specifying justmesh will make *cplreader* just read the mesh without any other data.
--degen	false	Allow degenerated element shapes. This option just applies to the ANSYS reader. If true all elements other than lines will degenerated version of quadrilaterals and hexahedrons.
--strict	false	Be strict. For the ANSYS reader this option means that *cplreader* will complain about holes in the node list and connectivity, i.e. if the user forgot to nummrg the entities in ANSYS. For the CFX reader this option means that if corrupt transient files have been found, the reader will complain and stop working.
--verbose	false	Be verbose If this parameter is set to true lots of helpful information will be output which is not necessery to have in most cases and would normally just clutter the shell.
--deffile		Definition file name. Only for CFX! Non-standard path to .def file for CFX.
--trntol	0.05	Ratio by which .trn files may differ. Ratio by which .trn files may differ before they are regarded corrupt. E.g.: Let's assume the mean size of .trn files is 1MB. By specifying –trntol 0.05 the files may vary in size from 0.95MB to 1.05MB before an error condition is assumed.
--lhsrc		Column of Lighthill source term in FASTEST result files (e.g. col1). If FASTEST already calculated the Lighthill source term, the column in the ASCII file can be specified using this parameter.
--vx		Column of x-velocity in FASTEST result files (e.g. col2). Specify column of x-velocity in FASTEST ASCII files.
--vy		Column of y-velocity in FASTEST result files (e.g. col3). Specify column of y-velocity in FASTEST ASCII files.
--vz		Column of z-velocity in FASTEST result files (e.g. col4). Specify column of z-velocity in FASTEST ASCII files.
--pres		Column of pressure in FASTEST result files (e.g. col5). Specify column of pressure in FASTEST ASCII files.
--reduceElementOrder	false	disregard high order nodes If this parameter is set, higher order noes of the elements will be disregarded for the calclation. Only reasonable for quadratic elements.
--doIntAverageCentre	false	averages the integration over nodes at centre

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Table 1 – continued from previous page

Parameter	Default Value	Short Description
		If this parameter is set acouRhsLoad is calculated by using the velocity vector at the centre of each element. May help with quadratic elements.
--doCalcMultiNodes	true	This flag needs to be set to remedy region interface artifacts. If multiple regions exist, nodes which are shared by both are stored multplie times, once for each region. But their acouRhsLoad is only calculated in each region seperatly and therefore get different values which need to be added. This flag should be set on otherwise artifacts at the region interface occur.

5 Theory

The contents of this section have been taken or are based upon in large parts from [14].

5.1 Introduction

A large amount of the total noise in our daily life is generated by turbulent flows (e.g. airplanes, cars, air conditioning systems, etc.). The physics behind the generation process is quite complicated and still not fully understood. The use of numerical simulation tools is one important way to analyze the generation of flow induced sound. Fig. 1 displays the general situation for the numerical simulation, we are concerned with, when performing computational aeroacoustics. Therein Ω_1 denotes the computational domain, where we have to solve for the flow field and the generation of sound, and Ω_2 the domain where the flow is zero, and where we are interested in the radiated sound.

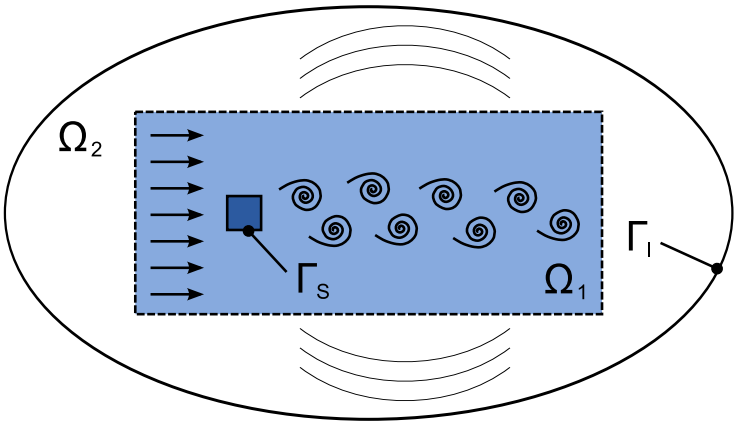


Figure 1: General setup of aeroacoustics

The surface Γ_S defines the interaction between structural mechanics and flow and Γ_I the boundary of the total computational domain. Now, the computation of the flow induced noise exhibits a lot of difficulties, where the most significant ones can be summarized as follows:

1. The scale discrepancy between flow pressure and acoustic pressure reaches values of 10^7.
2. The acoustic domain is in most cases large compared to the fluid flow domain.
3. The spatial discretization for the flow computation is much finer as compared to the discretization of the acoustic domain.
4. In many applications, the interface between flow and solid/elastic bodies leads to vibrations, which not only effect the flow, but also emits acoustic sound (vibrational acoustics).

Due to 1-3, a direct numerical simulation (DNS) of Navier Stokes equations, which are capable to describe both the flow as well as acoustic field, is restricted to very few practical cases (see e.g., [18]). Therefore, most approaches, published in the last 20 years, uses hybrid methods, e.g. [19], [5], [9]. Since we concentrate on a computational scheme, which allows the computation of the flow generated sound simultaneously with any flow induced vibrational sound (due to solid-fluid interactions), we base our scheme on Lighthill's analogy [15, 16]. This formulation will provide a direct coupling between flow and acoustic sound computation in the time domain and in addition will offer for both physical fields different discretization. At this point, we want to notice, that a similar approach can be found in [17], which however has just been investigated in the time-harmonic case.

The outline of this theory section is as follows: First of all, we give a review of Lighthill's analogy (Sec. 5.2) and its generalization. In Sec. 5.3 we will then concentrate on the weak formulation of Lighthill's analogy and its finite element (FE) formulation. Section 5.4 provides a description of our environment for computational aeroacoustics.

5.2 Lighthill's Analogy and its Extension

Let us start our derivation of Lighthill's analogy by introducing the two main equations of fluid dynamics: mass conservation and momentum conservation. The differential and coordinate-free formulation of the mass conservation (continuity) equation reads as follows

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_i}{\partial x_i} = 0\,. \tag{1}$$

In (Eq. (1)) ρ denotes the density of the fluid and v_i the i-th component of the flow velocity vector $\mathbf{v}$. The momentum conservation equation is expressed by

$$\rho \frac{\partial v_i}{\partial t} + \rho v_j \frac{\partial v_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} - \frac{\partial \tau_{ij}}{\partial x_j} = -\frac{\partial}{\partial x_j}\left(p\delta_{ij} + \tau_{ij}\right)\,, \tag{2}$$

where p denotes the total pressure within the flow, $[\tau]$ the viscous stress tensor and δ the Kronecker symbol. For Newtonian fluids, we can express $[\tau]$ by

$$\tau_{ij} = -\mu\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + \frac{2}{3}\mu\delta_{ij}\frac{\partial v_k}{\partial x_k} \tag{3}$$

with μ the dynamic viscosity.

In a first step, we multiply (Eq. (1)) by v_i (before we substitute the index i by j) and add the so obtained equation to (Eq. (2)). In addition, we substitute p in (Eq. (2)) by $p - p_0$ with p_0 the mean pressure. This operation is allowed, since the relation $\partial(p - p_0)/\partial x_j = \partial p/\partial x_j$ holds. Therewith, we obtain

$$\underbrace{\rho\frac{\partial v_i}{\partial t} + v_i\frac{\partial \rho}{\partial t}}_{\frac{\partial \rho v_i}{\partial t}} + \underbrace{\rho v_j\frac{\partial v_i}{\partial x_j} + v_i\frac{\partial \rho v_j}{\partial x_j}}_{\frac{\partial}{\partial x_j}(\rho v_i v_j)} = -\frac{\partial}{\partial x_j}((p - p_0)\delta_{ij} - \tau_{ij}) \tag{4}$$

By introducing the momentum flux tensor $[\tau^{\mathrm{I}}]$ by

$$\tau^{\mathrm{I}}_{ij} = \rho v_i v_j + (p - p_0)\delta_{ij} - \tau_{ij} \tag{5}$$

we may write (Eq. (4)) as

$$\frac{\partial \rho v_i}{\partial t} = -\frac{\partial \tau^{\mathrm{I}}_{ij}}{\partial x_j}\,. \tag{6}$$

For linear acoustics, the momentum flux tensor $[\tau^{\mathrm{I}}]$ reduces to

$$\tau^{0}_{ij} = (p - p_0)\delta_{ij} \tag{7}$$

and (Eq. (6)) reads as

$$\frac{\partial \rho v_i}{\partial t} + \frac{\partial}{\partial x_i}(p - p_0) = 0\,. \tag{8}$$

In addition, we define the acoustic pressure p', the acoustic density ρ' and its relation (just fulfilled for linear acoustics)

$$p' = p - p_0\,; \quad \rho' = \rho - \rho_0\,; \quad \frac{p'}{\rho'} = c_0^2 \tag{9}$$

with c_0 the speed of sound. Substituting $p - p_0$ in (Eq. (8)) by $c_0^2\rho'$ and applying $\partial/\partial x_i$ to this equation, results in

$$\frac{\partial}{\partial t}\frac{\partial}{\partial x_i}(\rho v_i) + \frac{\partial^2}{\partial x_i^2}(c_0^2\rho') = 0\,. \tag{10}$$

According to the mass conservation equation, we may rewrite (Eq. (10)) as

$$\frac{\partial^2 \rho'}{\partial t^2} + \frac{\partial^2}{\partial x_i^2}(c_0^2\rho') = \left(\frac{1}{c_0^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x_i^2}\right)(c_0^2\rho') = 0\,. \tag{11}$$

Since we have neglected the flow and there are no other excitations, no acoustic sound field will be generated, since the solution of (Eq. (11)) is $\rho' = \rho - \rho_0 = 0$.

Lighthill's analogy is based on the fact, that the sound generated by the flow in a real fluid is exactly equivalent to that produced in the ideal, linear acoustic field, forced by the stress distribution

$$
\begin{aligned}
L_{ij} &= \tau_{ij}^{\mathrm{I}} - \tau_{ij}^{0} \\
&= \rho v_i v_j + \left((p - p_0) - c_0^2(\rho - \rho_0)\right)\delta_{ij} - \tau_{ij}\,,
\end{aligned} \tag{12}
$$

where $[L]$ denotes the Lighthill stress tensor.

We can now rewrite (Eq. (6)) as the momentum equation for the ideal, linear acoustic medium subjected to the externally applied stress according to (Eq. (12))

$$
\frac{\partial \rho v_i}{\partial t} + \frac{\partial \tau_{ij}^{0}}{\partial x_j} = -\frac{\partial(\tau_{ij}^{\mathrm{I}} - \tau_{ij}^{0})}{\partial x_j} \tag{13}
$$

Therewith, we obtain

$$
\frac{\partial \rho v_i}{\partial t} + \frac{\partial}{\partial x_i}\left(c_0^2(\rho - \rho_0)\right) = -\frac{\partial L_{ij}}{\partial x_j}\,. \tag{14}
$$

Eliminating the momentum density ρv_i similarly as for the linear case, we obtain

$$
\left(\frac{1}{c_0^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x_i^2}\right)\left(c_0^2(\rho - \rho_0)\right) = \frac{\partial^2 L_{ij}}{\partial x_i \partial x_j}\,. \tag{15}
$$

The integral formulation of (Eq. (15)) is given by

$$
c_0^2(\rho - \rho_0)(\mathbf{x}, t) = \frac{1}{4\pi}\frac{\partial^2}{\partial x_i \partial x_j}\int\limits_{\Omega_1} \frac{\partial L_{ij}(\boldsymbol{\xi}\,, t - r/c_0)}{r}\,\mathrm{d}\xi\,\mathrm{d}\eta\,\mathrm{d}\zeta \tag{16}
$$

with the source coordinates $\boldsymbol{\xi} = (\xi, \eta, \zeta)^T$, the observation coordinates $\mathbf{x} = (x_1, x_2, x_3)^T$ and the distance vector $\mathbf{r} = (x_1 - \xi, x_2 - \eta, x_3 - \zeta)^T$.

The integral formulation given in (Eq. (16)) is no longer correct, if any solid and/or elastic bodies (in rest or moving) are present within the simulation domain. For such situations, the acoustic field has to fulfill the correct boundary condition and we no longer can use Green's function for free radiation. Now, the idea is, to derive an extended form of Lighthill's equation, which is defined in the whole simulation domain, and thus Green's function for free radiation can be applied to obtain an integral formulation [10].

We will assume a domain with a solid body, which is located within the considered domain Ω_1. The volume of the body is denoted by Ω_{B} and its surface by Γ_{S} (see Fig. 1). We describe the surface of the body with a function $f(\mathbf{x}, t)$

$$
f(\mathbf{x}, t) = \begin{cases} f < 0 & \text{for } \mathbf{x} \in \Omega_{\mathrm{B}} \\ f > 0 & \text{for } \mathbf{x} \in \Omega_1 \setminus \Omega_{\mathrm{B}} \\ f = 0 & \text{for } \mathbf{x} \text{ on } \Gamma_{\mathrm{S}} \end{cases} \tag{17}
$$

e.g., a breathing sphere: $f(\mathbf{x}, t) = |\mathbf{x} - \mathbf{x}_o|^2 - R_0^2(t)$. Additionally, we introduce the Heaviside function $H(f(\mathbf{x}, t))$

$$
H(f(\mathbf{x}, t)) = \begin{cases} 1 & \text{for } \mathbf{x} \in \Omega_1 \setminus \Omega_{\mathrm{B}} \\ 0 & \text{for } \mathbf{x} \in \Omega_{\mathrm{B}} \end{cases}\,. \tag{18}
$$

First of all, we multiply the continuity equation (see (Eq. (1))) with $H(f)$

$$
\left(\frac{\partial \rho'}{\partial t} + \frac{\partial}{\partial x_i}(\rho v_i)\right) H(f) = 0 \tag{19}
$$

and rewrite it in the following form

$$
\frac{\partial}{\partial t}\left(\rho' H(f)\right) - \rho'\frac{\partial H(f)}{\partial t} + \frac{\partial}{\partial t}\left((\rho v_i) H(f)\right) - (\rho v_i)\frac{\partial H(f)}{\partial x_i} = 0\,. \tag{20}
$$

The Heaviside function $H(f)$ fulfills two important properties [11]

$$
\frac{\partial H(f)}{\partial x_i} = \frac{\partial f}{\partial x_i}\delta(f) \tag{21}
$$

$$
\frac{\partial H(f)}{\partial t} = \delta(f)\frac{\partial f}{\partial t} = -v_i^{\mathrm{B}}\frac{\partial f}{\partial x_i}\delta(f) \tag{22}
$$

with v_i^{B} the velocity of the solid body in Ω_1. Using (Eq. (21)) and (Eq. (22)) simplifies (Eq. (20)) to

$$\frac{\partial(\rho' H(f))}{\partial t} + \frac{\partial(\rho v_i H(f))}{\partial x_i} = \left(\rho(v_i - v_i^{\mathrm{B}}) + \rho_0 v_i^{\mathrm{B}}\right) \frac{\partial f}{\partial x_i}\delta(f)\,. \tag{23}$$

This equation is an extension of the continuity equation, which is fulfilled all over Ω. As can be seen, the right hand side of (Eq. (23)) is just on Γ_{S} different from zero and the left side is zero within Ω_{B}. Applying similar operations to the momentum conservation equation, we obtain

$$\begin{aligned} \frac{\partial}{\partial t}\left(\rho v_i H(f)\right) \quad &+ \quad \frac{\partial}{\partial x_j}\left(\rho v_i v_j + p'\delta_{ij} - \tau_{ij}\right) H(f) \\ &= \quad \left(\rho v_i (v_j - v_j^{\mathrm{B}}) + p'\delta_{ij} - \tau_{ij}\right) \frac{\partial f}{\partial x_j}\delta(f)\,. \end{aligned} \tag{24}$$

Using the definition of the Lighthill tensor L_{ij} (see (Eq. (12))), we may write (Eq. (24)) as

$$\begin{aligned} \frac{\partial}{\partial t}\left(\rho v_i H(f)\right) \quad &+ \quad \frac{\partial}{\partial x_i}(c_0^2 \rho' H(f)) \\ &= \quad -\frac{\partial L_{ij} H(f)}{\partial x_j} + \left(\rho v_i (v_j - v_j^{\mathrm{B}}) + p'\delta_{ij} - \tau_{ij}\right) \frac{\partial f}{\partial x_j}\delta(f)\,. \end{aligned} \tag{25}$$

Furthermore, we rewrite (Eq. (23)) in the following form

$$\frac{\partial(\rho v_i H(f))}{\partial t} = \left(\rho(v_i - v_i^{\mathrm{B}}) + \rho_0 v_i^{\mathrm{B}}\right) \frac{\partial f}{\partial x_i}\delta(f) - \frac{\partial(\rho' H(f))}{\partial x_i}\,. \tag{26}$$

Applying $\partial/\partial x_i$ to (Eq. (25)) and using the relation according to (Eq. (26)), we arrive at the extended form of Lighthill's equation

$$\begin{aligned} &\left(\frac{1}{c_0^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x_i^2}\right)\left(c_0^2 \rho'\, H(f)\right) \\ = \quad &\frac{\partial^2}{\partial x_i \partial x_j}\left(L_{ij} H(f)\right) - \frac{\partial}{\partial x_i}\left(\left(\rho v_i (v_j - v_j^{\mathrm{B}}) + p'\,\delta_{ij} - \tau_{ij}\right) \frac{\partial f}{\partial x_j}\delta(f)\right) \\ &+ \frac{\partial}{\partial t}\left(\left(\rho(v_i - v_i^{\mathrm{B}}) + \rho_0 v_i^{\mathrm{B}}\right) \frac{\partial f}{\partial x_i}\delta(f)\right)\,. \end{aligned} \tag{27}$$

Since the extended form of Lighthill's equation is valid throughout the whole domain Ω, we can use Green's function for free radiation to obtain a solution. Therefore, the integral form, which is known as the *Ffowcs Williams and Hawkings equation* [10], reads as

$$\begin{aligned} \rho'\, H(f) = \frac{\partial^2}{\partial x_i \partial x_j}\int\limits_{\Omega_1} \frac{\partial L_{ij}}{4\pi\, r}\, d\Omega \quad &- \quad \frac{\partial}{\partial x_i}\oint\limits_{\Gamma_{\mathrm{S}}} \frac{\left(\rho(v_i - v_i^{\mathrm{B}}) + p'\,\delta_{ij} - \tau_{ij}\right)}{4\pi\, r}\, \mathrm{d}\Gamma \\ &+ \quad \frac{\partial}{\partial t}\oint\limits_{\Gamma_{\mathrm{S}}} \frac{\left(\rho(v_i - v_i^{\mathrm{B}}) + \rho v_i^{\mathrm{B}}\right)}{4\pi\, r}\, \mathrm{d}\Gamma\,. \end{aligned} \tag{28}$$

5.3 Finite Element Formulation

We will perform a volume discretization of Lighthill's equation (see (Eq. (15))) by applying the finite element method (FEM). Therewith, any solid/elastic body will be implicitly taken into account, and there is no need to use the extended form of Lighthill's equation as given by (Eq. (27)).

Let us start at the strong formulation of (Eq. (15)), which reads as follows

$$\begin{aligned}
\text{Given:} \quad & L_{ij} && : \Omega \times (0,T) \to \mathbb{R} \\
& c_0 && : \Omega \to \mathbb{R} \\
& \rho'^0 && : \Omega \to \mathbb{R} \\
& \dot{\rho}'^0 && : \Omega \to \mathbb{R} \\
\text{Find:} \quad & \rho' && : \bar{\Omega} \times [0,T] \to \mathbb{R}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 \rho'}{\partial t^2} - c_0^2 \frac{\partial^2 \rho'}{\partial x_i^2} &= \frac{\partial^2 L_{ij}}{\partial x_i \partial x_j} \\
\rho'(\mathbf{r},0) &= \rho'^0 \,, \; \mathbf{r} \in \Omega \\
\dot{\rho}'(\mathbf{r},0) &= \dot{\rho}'^0 \,, \; \mathbf{r} \in \Omega
\end{aligned} \tag{29}$$

At this stage we assume, that the Lighthill tensor $[L]$ is a known quantity obtained by a flow computation using a large eddy simulation (LES) or other appropriate type of CFD scheme. In a first step, we multiply (Eq. (29)) by an appropriate test function w and integrate over the whole domain Ω (corresponding in Fig. 1 to $\Omega_1 \cup \Omega_2$)

$$\int_\Omega w \left(\frac{\partial^2 \rho'}{\partial t^2} - c_0^2 \frac{\partial^2 \rho'}{\partial x_i^2} - \frac{\partial^2 L_{ij}}{\partial x_i \partial x_j} \right) \mathrm{d}\Omega = 0 \,. \tag{30}$$

Now, we apply Green's integral theorem to the second spatial derivative of ρ' as well as L_{ij}. This operation will result in the following relations

$$\int_\Omega c_0^2 \, w \frac{\partial^2 \rho'}{\partial x_i^2} \, \mathrm{d}\Omega = \int_{\Gamma_\mathrm{S} \cup \Gamma_\mathrm{I}} w \frac{\partial \rho'}{\partial n} \, \mathrm{d}\Gamma - \int_\Omega c_0^2 \, \frac{\partial w}{\partial x_i} \frac{\partial \rho'}{\partial x_i} \, \mathrm{d}\Omega \tag{31}$$

$$\int_\Omega w \, \frac{\partial^2 L_{ij}}{\partial x_i \partial x_j} \, \mathrm{d}\Omega = \int_{\Gamma_\mathrm{S}} w \frac{\partial L_{ij}}{\partial x_j} n_i \, \mathrm{d}\Gamma - \int_\Omega \frac{\partial w}{\partial x_i} \frac{\partial L_{ij}}{\partial x_j} \, \mathrm{d}\Omega \,. \tag{32}$$

We want to emphasize, that the boundary integral (Eq. (32)) is just over the surface Γ_S of any solid/elastic body, whereas in (Eq. (31)) we have to integrate over Γ_S as well as over Γ_I, which limits the computational domain (see Fig. 1).

Now we can substitute $\partial L_{ij}/\partial x_j$ within the first term on the right hand side of (Eq. (32)) by (Eq. (14)) and obtain

$$\int_{\Gamma_\mathrm{S}} w \frac{\partial L_{ij}}{\partial x_j} n_i \, \mathrm{d}\Gamma = \int_{\Gamma_\mathrm{S}} w \left(-\frac{\partial \rho v_i}{\partial t} - \frac{\partial}{\partial x_i}(c_0^2 \rho') \right) n_i \, \mathrm{d}\Gamma \,. \tag{33}$$

Since on a solid surface $v_i n_i = 0$ is fulfilled, we see, that the surface integral term over Γ_S reduces to

$$\int_{\Gamma_\mathrm{S}} w \frac{\partial L_{ij}}{\partial x_j} n_i \, \mathrm{d}\Gamma = - \int_{\Gamma_\mathrm{S}} c_0^2 \, w \, \frac{\partial \rho'}{\partial n} \, \mathrm{d}\Gamma \,. \tag{34}$$

Therewith, we can rewrite (Eq. (30)) as

$$\begin{aligned}
\int_\Omega w \frac{\partial^2 \rho'}{\partial t^2} \quad &+ \quad \int_\Omega c_0^2 \, \frac{\partial w}{\partial x_i} \frac{\partial \rho'}{\partial x_i} \, \mathrm{d}\Omega \; - \int_{\Gamma_\mathrm{S} \cup \Gamma_\mathrm{I}} c_0^2 \, w \frac{\partial \rho'}{\partial n} \, \mathrm{d}\Gamma \\
&= \quad - \int_\Omega \frac{\partial w}{\partial x_i} \frac{\partial L_{ij}}{\partial x_j} \, \mathrm{d}\Omega - \int_{\Gamma_\mathrm{S}} c_0^2 \, w \frac{\partial \rho'}{\partial n} \, \mathrm{d}\Gamma \,.
\end{aligned} \tag{35}$$

Combining the surface integrals results in a single boundary integral just performed over the outer boundary Γ_I, on which we apply absorbing boundary conditions of first order according to [7]. Therewith, we utilize the relation

$$c_0 \frac{\partial \rho'}{\partial n} = \frac{\partial \rho'}{\partial t} \tag{36}$$

and arrive at the weak form of(Eq. (29)): Find $\rho' \in H^1$ such that

$$\int_\Omega w\frac{\partial^2\rho'}{\partial t^2} + \int_\Omega c_0^2\,\frac{\partial w}{\partial x_i}\,\frac{\partial \rho'}{\partial x_i}\;\mathrm{d}\Omega \;-\; \int_{\Gamma_\mathrm{I}} c_0^2\, w\frac{\partial \rho'}{\partial n}\;\mathrm{d}\Gamma = -\int_\Omega \frac{\partial w}{\partial x_i}\frac{\partial L_{ij}}{\partial x_j}\;\mathrm{d}\Omega \tag{37}$$

for any $w \in H^1$. With H^1 we denote the Sobolev space defined as (see e.g. [4])

$$H^1 = \{u \in L^2 | \partial u/\partial x_i \in L^2\} \tag{38}$$

and L^2 the space of square integrable functions.

Using standard nodal finite elements, we approximate the continuous acoustic density ρ' as well as the test function w by

$$\rho' \;\approx\; {\rho'}^h = \sum_{a=1}^{n_\mathrm{eq}} N_a \rho'_a \tag{39}$$

$$w \;\approx\; w^h = \sum_{a=1}^{n_\mathrm{eq}} N_a w_a\,. \tag{40}$$

Thus, (Eq. (37)) is transformed to the following semidiscrete Galerkin formulation

$$\mathbf{M}\underline{\ddot{\rho}'}_{n+1} + \mathbf{C}\underline{\dot{\rho}'}_{n+1} + \mathbf{K}\underline{\rho'}_{n+1} = \underline{f}_{n+1} \tag{41}$$

with $\underline{\ddot{\rho}'} = \partial^2\rho_\sim/\partial t^2$, $\underline{\dot{\rho}'} = \partial\rho_\sim/\partial t$, $\underline{\rho'}$ the nodal unknowns of the acoustic density and n the time step counter. The matrices as well as right hand side vector compute as follows:

$$\mathbf{M} \;=\; \bigwedge_{e=1}^{n_\mathrm{e}} \mathbf{m}^e \;;\; \mathbf{m}^e = [m_{pq}] \;;\; m_{pq} = \int_{\Omega^e} N_p N_q\;\mathrm{d}\Omega \tag{42}$$

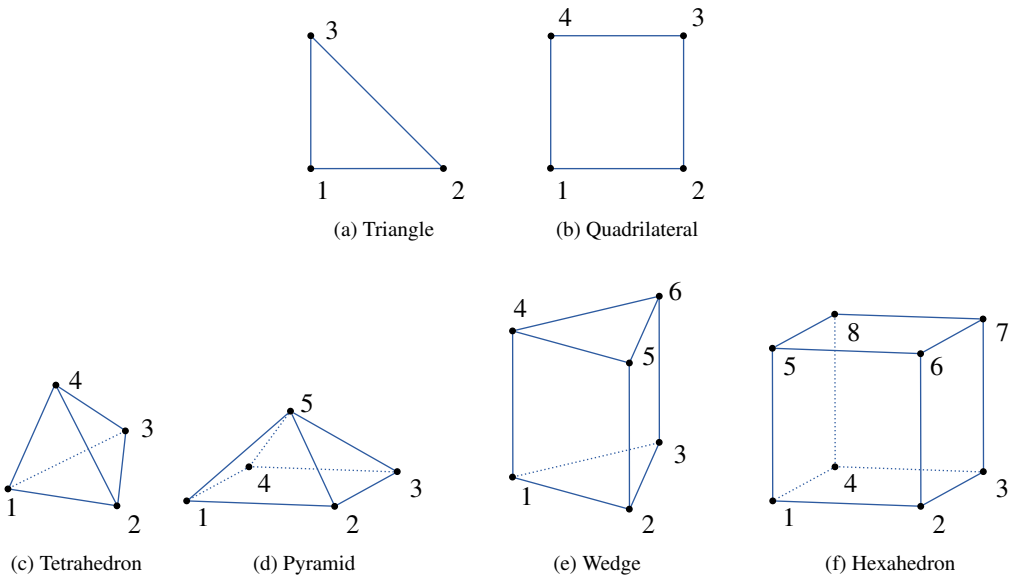
$$\mathbf{C} \;=\; \bigwedge_{e=1}^{n_{\Gamma_\mathrm{I}}} \mathbf{c}^e \;;\; \mathbf{c}^e = [c_{pq}] \;;\; c_{pq} = \int_{\Gamma_\mathrm{I}^e} c_0\, N_p N_q\;\mathrm{d}\Gamma \tag{43}$$

$$\mathbf{K} \;=\; \bigwedge_{e=1}^{n_\mathrm{e}} \mathbf{k}^e \;;\; \mathbf{k}^e = [k_{pq}] \;;\; k_{pq} = \int_{\Omega^e} c^2\,\frac{\partial N_p}{\partial x_i}\,\frac{\partial N_q}{\partial x_i}\;\mathrm{d}\Omega \tag{44}$$

$$\underline{f}_{n+1} \;=\; \bigwedge_{e=1}^{n_\mathrm{e}} \underline{f}^e \;;\; \underline{f}^e = [f_p] \;;\; f_p = \int_{\Omega^e} \frac{\partial N_p}{\partial x_i}\frac{\partial L_{ij}^{n+1}}{\partial x_j}\;\mathrm{d}\Omega\,. \tag{45}$$

In the above equations n_e is the number of finite elements, n_{Γ_I} the number of finite surface elements along the outer boundary Γ_I and $\bigwedge$ the finite element assembly operator.

The following element types with their corresponding nodal basis functions are supported for the calculation of acoustic sources.



(a) Triangle (b) Quadrilateral

(c) Tetrahedron (d) Pyramid (e) Wedge (f) Hexahedron

Figure 2: Supported Element Types

Element type	**Reference points**				**Nodal basis functions**
	Node i	ξ	η	ζ	
Triangle	1	0	0	0	$N_1 = 1 - \xi - \eta$
	2	1	0	0	$N_2 = \xi$
	3	0	1	0	$N_3 = \eta$
Quadrilateral	1	-1	-1	0	$N_i = \frac{1}{4}(1 + \xi_i\xi)(1 + \eta_i\eta)$
	2	1	-1	0	
	3	1	1	0	
	4	-1	1	0	
Tetrahedron	1	0	0	0	$N_1 = 1 - \xi - \eta - \zeta$
	2	1	0	0	$N_2 = \xi$
	3	0	1	0	$N_3 = \eta$
	4	0	0	1	$N_4 = \zeta$
Pyramid	1	1	0	0	
	2	-1	1	0	
	3	-1	-1	0	
	4	1	-1	0	
	5	0	0	1	
Wedge	1	0	0	-1	$N_1 = (1 - \xi - \eta)(1 + \zeta_i\zeta)$
	2	1	0	-1	$N_2 = \xi(1 + \zeta_i\zeta)$
	3	0	1	-1	$N_3 = \eta(1 + \zeta_i\zeta)$
	4	0	0	1	$N_4 = (1 - \xi - \eta)(1 + \zeta_i\zeta)$
	5	1	0	1	$N_5 = \xi(1 + \zeta_i\zeta)$
	6	0	1	1	$N_6 = \eta(1 + \zeta_i\zeta)$
Hexahedron	1	-1	-1	-1	$N_i = \frac{1}{8}(1 + \xi_i\xi)(1 + \eta_i\eta)(1 + \zeta_i\zeta)$
	2	1	-1	-1	
	3	1	1	-1	
	4	-1	1	-1	
	5	-1	-1	1	
	6	1	-1	1	
	7	1	1	1	
	8	-1	1	1	

Table 2: Element types with their corresponding node coordinates and nodal basis functions

The time discretization is performed by applying a standard Newmark algorithm, which results in the following time-stepping scheme (effective mass matrix formulation) [12]:

- Perform predictor step:

$$\underline{\tilde{\rho}'} = \underline{\rho'}_n + \Delta t\, \underline{\dot{\rho}'}_n + \frac{\Delta t^2}{2}(1-2\beta_{\mathrm{H}})\, \underline{\ddot{\rho}'}_n \tag{46}$$

$$\underline{\tilde{\dot{\rho}}'} = \underline{\dot{\rho}'}^n + (1-\gamma_{\mathrm{H}})\Delta t\, \underline{\ddot{\rho}'}_n\,. \tag{47}$$

- Solve algebraic system of equations:

$$\mathbf{M}^*\, \underline{\ddot{\rho}'}_{n+1} = \underline{f}_{n+1} - \mathbf{K}\underline{\tilde{\rho}'} - \mathbf{C}\underline{\tilde{\dot{\rho}}'} \tag{48}$$

$$\mathbf{M}^* = \mathbf{M} + \gamma_{\mathrm{H}}\Delta t\, \mathbf{C} + \beta \Delta t^2\, \mathbf{K}\,. \tag{49}$$

- Perform corrector step:

$$\underline{\rho'}_{n+1} = \underline{\tilde{\rho}'} + \beta_{\mathrm{H}}\Delta t^2\, \underline{\ddot{\rho}'}_{n+1} \tag{50}$$

$$\underline{\dot{\rho}'}_{n+1} = \underline{\dot{\tilde{\rho}}'} + \gamma_{\mathrm{H}}\Delta t\, \underline{\ddot{\rho}'}_{n+1}\,. \tag{51}$$

In (Eq. (46)) - (Eq. (51)) Δt denotes the time step value and β_{H}, γ_{H} integration parameters, which are set to 0.25 and 0.5 to obtain a fully implicit time integration scheme.

5.4 Calculation Scheme

We propose a coupling between flow and acoustic computations, which allows to choose an optimal discretization for each physical field. The flow may be computed by the codes *ANSYS CFX*, *OpenFOAM*, *FASTEST-3D* [6]. *FASTEST-3D* uses a fully conservative second-order finite volume space discretization and a pressure correction method of the SIMPLE type for the iterative coupling of velocity and pressure. *ANSYS CFX* uses the Scale Adaptive Simulation (SAS) scheme. *OpenFOAM* is an open source CFD solver.

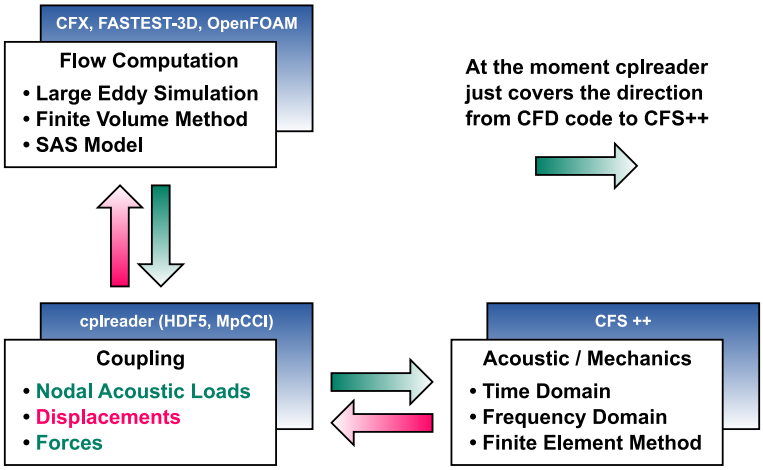


Figure 3: Coupling Scheme

For time discretization an implicit second-order scheme is employed, while a non-linear multigrid scheme, in which the pressure correction method acts as a smoother on the different grid levels, is used for convergence acceleration. The turbulent flow is computed by an appropriate CFD scheme like the large eddy simulation (LES) applying the Smagorinsky model or the SAS model of *ANSYS CFX*. The concept of block structured grids is employed in *FASTEST-3D* to handle complex geometries and for the ease of parallelization. The parallel implementation is based on grid partitioning with automatic load balancing and follows the message-passing concept, ensuring a high degree of portability.

The computation of the generated acoustic noise is performed by the finite element method as discussed in Sec. 5.3 and has been implemented in CFS++ (Coupled Field Simulation) [13]. In order to minimize the amount of data transfer and to achieve a higher accuracy, we already compute the right hand side $\underline{f}^{n+1}$ (see Eq. (45)) on the much finer flow grid and obtain so called acoustic nodal loads. This is performed as a post-processing step of the CFD computation by cplreader. Within each time step, these scalar values are interpolated to the acoustic grid by using a customized conservative interpolation algorithm in *CFS++* or MpCCI - the Mesh-based parallel Code Coupling Interface [2] (see Fig. 3). In doing so the weights for the distribution of the CFD quantity to the nodes in the acoustic cell are obtained by evaluating the basis functions defined in Tab. 2 at the position of the CFD node. More technical details on this issue may be found in Section 6.4.2 in [3] which is also available as part of the cplreader documentation.

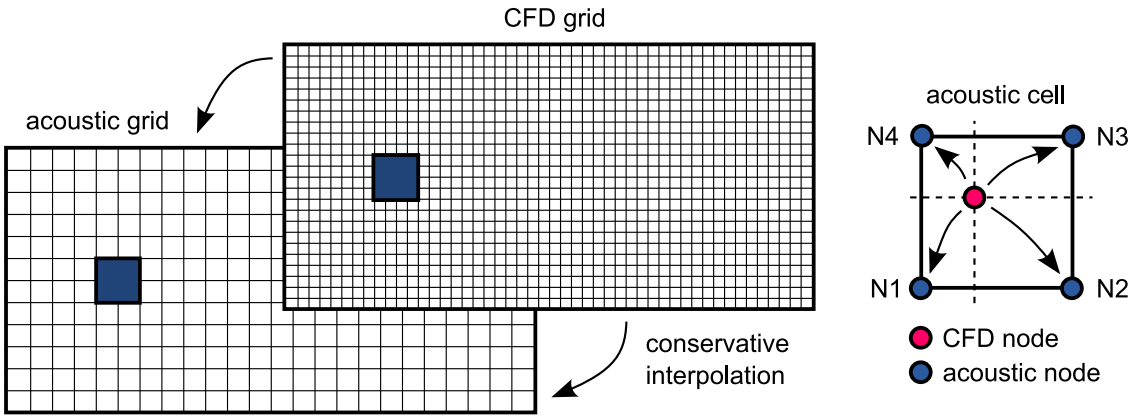


Figure 4: Conservative interpolation of source terms from CFD to acoustic grid.

The data transfer has to be performed within the whole computational domain of the flow (corresponds to Ω_1 in Fig. 1). In order to obtain a correct interpolation, the acoustic nodal loads have then to be interpolated by a *conservative technique*, since they correspond to an integral quantity.

Therewith, the developed scheme allows a direct coupling in time domain and provides the acoustic sound field not only in the far field but also in the region of the flow. Furthermore, we can directly investigate the acoustic source terms in the flow region and the scheme is well suited for complex geometries and fluid structure interaction. By applying a discrete Fourier transform to the acoustic nodal loads, it is also possible to perform analyses in the frequency domain.

In addition to the full 3D coupling, we have implemented a 2D coupling from the flow to the acoustic computation. Therewith, we just compute the acoustic nodal loads on a predefined 2D slice and perform the acoustic field computation only within a cross section. It is also possible to slice a 3D CFD dataset by interpolating to a 2D acoustic grid. In both cases the acoustic nodal loads have been previously computed and written to a *HDF5* file by cplreader. This file has then been read and interpolated by *CFS++* (cf. Fig. 4). This technique allows us to write the computed acoustic nodal loads to a file and perform acoustic simulations independent of the flow computation.

6 CFD Data Generators

The first purpose of cplreader when started by Max Escobar was to just read velocity fields generated by FASTEST-3D and to provide this data to CFS++ via an MpCCI interface. Since then cplreader has evolved from just a simple reader to an intermediate program which can also calculate acoustic source terms.

These source terms can also be generated by cplreader from analytic expressions. This has however not been implemented yet. Possible extensions for cplreader include:

- Max Escobars vortex core model from AcouFlowNoise.cc
- Source term reader for AcouCombustion

7 CFD Data Readers

7.1 FASTEST-3D

7.1.1 Required File Layout

```
./name
|-- name_01.coord
|-- name_01.topo
|-- name_01_0001.dat
|-- name_01_0002.dat
|-- name_02.coord
|-- name_02.topo
|-- name_02_0001.dat
`-- name_02_0002.dat
```

7.1.2 Special Options

`--lhsrc` Column of Lighthill source term in FASTEST result files (e.g. col1). If FASTEST already calculated the Lighthill source term, the column in the ASCII file can be specified using this parameter.

`--vx, --vy, --vz` Columns of x, y, z-velocities in FASTEST result files (e.g. col2).

`--pres` Column of hydrodynamic pressure in FASTEST result files (e.g. col5).

7.1.3 ASCII File Format

`.coord`-File Format

```
 1   1   2   6
  0.  0.  0.0
  0.  0.03125  0.0
  0.  0.09375  0.0
  0.03125  0.  0.0
  0.03125  0.03125  0.0
  0.03125  0.09375  0.0
```

`.topo`-File Format

```
  1 1 2 2 4
 4 5 2 1
 5 6 3 2
```

`.dat`-File Format

```
       1
 -5.30794066E-06
 -1.39465176E-05
 -2.06977642E-05
 -1.94136509E-05
 -3.94903112E-05
 -7.02913899E-05
```

7.1.4 Examples

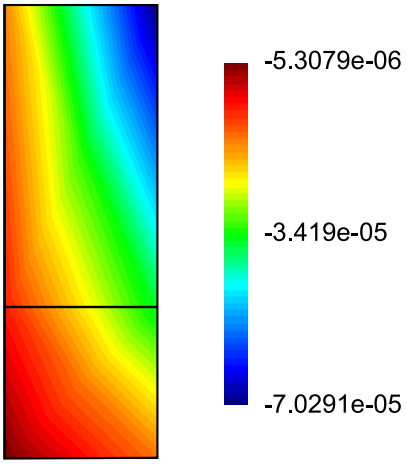


Figure 5: Visualization of dataset defined by above files.

7.2 ANSYS CFX

There exist some known problems for reading CFX 12.x files. Please refer to the description of the files section for a workaround.

7.2.1 Required File Layout

```
./name/
|-- name
|   |-- 35445.trn
|   `-- 35446.trn
|-- name.def
`-- name.res
```

All entities 'name' may also be symbolic links.

7.2.2 Description of the Files

.def The .def file contains the mesh with informations on its partitioning.

.res The .res file acts as a result file as well as restart file for CFX runs. It is the only file needed to restart a CFX computation. This means that, if the frequency resolution of the harmonic source term field is too low due to too few .trn files, one can always produce additional .trn steps by restarting the CFX computation from the .res file. It also contains the mesh plus the simulation description for the CFX run in the COMMANDS dataset. The COMMANDS dataset is comparable to our .xml file in addition to our material file. It contains such information as the time step size, which time steps get written to .trn files (e.g. only every second step), a string dataset with the file names of all .trn files, material properties, boundary conditions, etc. For later reference the COMMANDS dataset gets written to the UserData group of the resulting HDF5 file. Since the .res file also contains the mesh, one may create a symbolic link name.def to name.res, if a .def file is missing.

It has been found however that .res files produced with CFX version 12.x can cause problems with cplreader (Trac ticket https://lse17.e-technik.uni-erlangen.de:2001/trac/ticket/165). If this is the case one can get rid of the problem by only reading the mesh from the .def file. The .res file should be deleted or renamed for this purpose. The list of .trn files is obtained by cplreader by inspecting the contents of 'name' subdirectory instead of reading the corresponding dataset from the .res file in this case. Please note that the COMMANDS dataset will not be available in the HDF5 file, if no .res is present. Therefore the information about time step size has to be obtained in a different manner.

.trn The .trn files contain the actual field data for the time steps. The cplreader tries to read as much information as possible from the .trn files if the '–outputfields all' option is specified. In addition to the velocity field which is required for the calculation of the Lighthill source term, this includes the turbulent kinetic energy and density fields. There will however be no crash, if these fields are not present. If no –outputfields or no –calcsrc parameters are specified nothing but the mesh from the .def file will be read at all.

7.2.3 Special Options

`--type` The type of input files have to be set to 'CFX'.

`--deffile` You can specify different '.def' file than the one given by 'name.def'.

`--trntol` Percentage the file size of transient files may differ from the average file size of all transient files. If a transient file is smaller than average size * (1 - trntol) the corresponding time step will be overwritten by the previous one. This option has been introduced to be more robust against damaged .trn files, which may result during faulty data transfers.

7.2.4 Logging

Opening the root *HDF5*file ('name'.h5) one can find in /UserData/CFX_COMMANDS information about the CFX run which produced the .res file. This includes timestep size, material properties, boundary conditions, turbulence model, etc. The string dataset /UserData/TRN_TO_STEP_MAP contains information about which .trn file was mapped to which timestep. This is useful if corrupt .trn files have been encountered which got replaced by the last previous non-corrupt .trn file.

7.2.5 Examples

Let us assume, that we have got a directory layout like the following:

```
.
|-- fluid_old.def
|-- fluid_001
|   |-- 0.trn          1000 KB
|   |-- 1.trn          1000 KB
|   |-- 2.trn           950 KB
|   |-- 3.trn          1000 KB
|   |-- 4.trn          1000 KB
|   |-- 5.trn          1000 KB
|   |-- 6.trn           900 KB
|   |-- 7.trn          1000 KB
|   |-- 8.trn          1000 KB
|   |-- 9.trn          1000 KB
|   |-- 10.trn         1000 KB
|     ...
`-- fluid_001.res
```

We first have to rename or relink all entities 'fluid_001' to 'fluid' so that they have consistent names. We do not have a 'fluid.def' where the settings of the CFX simulation are saved. But let's also assume you have a 'fluid_old.def' which has the same grid information in it. You may also use this '.def' file for reading the grid information. Now you could start reading in the data by issuing:

```
cplreader --type CFX --name fluid --numsteps 11 --timestep 1e-4 \
          --deffile fluid_old.def --trntol 0.05 --outputfields 'all'
```

This would read all transient files except 6.trn which would be replaced by 5.trn. You can even replace the .def file by the .res file since the grid is also saved in the .res file. The transient files get searched by cplreader in this case.

Please also refer to Sections 7.9 and 7.8 for further options for transferring CFD data from CFX to cplreader.

7.2.6 CFX File Format Internals

Since the file format used by CFX is closed source, most of its internal data stuctures had to be found by reverse engineering. This section summarizes, which dataset contains relevant information.

CFX files are binary files, which can be read by `Fortran` functions provided by ANSYS. These functions take as the most important arguments the dataset name ("what") and the location ("where"). While a dataset's name determines the meaning of the dataset's contents, the location determines the part of the simulation that the dataset's contents refers to. Global information is tagged with the location `EVERY`. The geometry is referred to by the location `ZN1` ("zone 1"). The blocks of the mesh are called *element sets*, located by `ZN1/ESx` (`x` must be replaced by the number of the element set). Boundary cells are grouped in *face sets*, located by `ZN1/FSx`. Table 3 contains an (incomplete) overview of the datasets in a .def file.

Table 3: Meaning of datasets in CFX .def files

What	Where	Description	Datatype
G/CLBSF	ZN1	Labels (names) of surfaces	*char
G/CLBVL	ZN1	Labels (names) of volumes	*char
G/CNAMCFX	EVERY	Name and path of .cfx file	*char
G/COMMANDS	EVERY	Definition of simulation run	*char
G/CRDVX	ZN1/VX	Coordinates of vertices	double[]
G/CRELNO	EVERY	CFX release number	*char
G/ILTPES	ZN1	Element type of element set 4: Tetrahedra 5: Prisms 6: Hexahedra 7: Pyramids	int
G/KVXPE	ZN1/ESx	Vertices per element in element set x	int[]
G/KELPE	ZN1/ESx	Element indexes in element set x	int[]
G/KELPF	ZN1/FSx	Surface elements in face set x Given by tuple (element number, face index)	int[]
G/NBCP	EVERY	Number of boundary conditions	int
G/NDIMS	ZN1	Dimension	int
G/NEL	ZN1	Number of elements	int
G/NELES	ZN1	Number of elements per set	int[]
G/NES	ZN1	Number of element sets (blocks)	int
G/NFC	ZN1	Number of faces (surface elements)	int
G/NFCFS	ZN1	Number of faces per face set (surface region)	int[]
G/NFS	EVERY ZN1	Number of face sets (surface regions)	int
G/NVX	ZN1	Number of vertices	int

7.3 CFX_EXPORT

7.3.1 Required File Layout

```
./name/
|--results_1.csv
`--results_2.csv
```

7.3.2 Required File Format

Data is stored in *groups*, that must begin with a line like `[Name]`. Group names are case sensitive. Groups not mentioned here are ignored. Lines that begin with `#` are ignored.

`Data` group This group contains the coordinates of nodes as well as the flow data per node. The first line contains the column headers separated by comma. Each column header consists of a name of a field followed by its unit enclosed in brackets, e.g. "`Pressure [ Pa ]`". Table 4 shows the recognized column headers. All following lines up to the next group (or end of file) provide the data using one line per node and values separated by comma.

Table 4: Data fields supported by CFX_EXPORT reader

Field name	Unit	Description
`Node Number`		non-negative unique integer number identifying the node
`X`	`m`	x coordinate of the node (required)
`Y`	`m`	y coordinate of the node (required)
`Z`	`m`	z coordinate of the node
`Density`	`kg m^-3`	
`Pressure`	`Pa`	
`Velocity u`	`m s^-1`	x component of velocity in computational frame
`Velocity v`	`m s^-1`	y component of velocity in computational frame
`Velocity w`	`m s^-1`	z component of velocity in computational frame

`Faces` group This group contains the connectivity of nodes. There must be no column header line. Each line provides the node numbers (as defined in the `Data` group) of one finite element, separated by comma. Elements with 3, 4, 5, or 6 nodes are allowed. Elements with 5 or 6 nodes will be split into triangles.

Although every data file may contain different node coordinates and different connectivity, the mesh is only read from the first data file.

7.3.3 Examples

```
[Data]
Node Number, X [ m ], Y [ m ], Velocity u [ m s^-1 ], Velocity v [ m s^-1 ]
0, 0.000e+00, 0.000e+00, 4.200e+01, 1.000e+01
1, 0.000e+00, 3.125e-02, 4.200e+01, 1.200e+01
2, 0.000e+00, 9.375e-02, 4.200e+01, 1.100e+01
3, 3.125e-02, 0.000e+00, 4.200e+01, 0.000e+00
4, 3.125e-02, 3.125e-02, 4.200e+01, 2.000e+00
5, 3.125e-02, 9.375e-02, 4.200e+01, 1.000e+00

[Faces]
3, 4, 1, 0
4, 5, 2, 1
```

7.4 OpenFOAM

7.4.1 Required File Layout

```
./name/
|-- 0
|   |-- U
|   |-- motionDiff
|   |-- motionU
|   `-- p
|-- 0.01
|   |-- U.gz
|   |-- motionU.gz
|   |-- p.gz
|   `-- polyMesh
|-- 0.02
|   |-- U.gz
|   |-- motionU.gz
|   |-- p.gz
|   `-- polyMesh
|-- constant
|-- log
`-- system
```

To inspect the OpenFOAM data before converting it using cplreader you can open it in the latest version of ParaView (3.8.x). The file to be loaded is ./name/system/controlDict. Choose the OpenFOAM reader from the list of available readers.

The OpenFOAM reader in the trunk version of cplreader does not support the latest version of OpenFOAM. For a newer reader developed by Patrick Lorenz and Jens Grabinger checkout https://lse17.e-technik.uni-erlangen.de:2001/svn/CFS++/branches/starccm.

7.4.2 Special Options

7.4.3 Logging

Opening the root *HDF5*file ('name'.h5) one can find in /UserData all files from the directory structure of OpenFOAM which are necessary to restart the OpenFOAM simulation. This includes the files in the following directories:

```
./name/
|-- 0
|-- constant
|-- log
`-- system
```

Informations like timestep size, mesh, turbulence model, etc. should be available from these files.

7.4.4 Examples

For examples on how to start cplreader with OpenFOAM data you can ask Stefan Zörner or look at the cplreader_performance_suite. You may have a look at /home/lse24/lse/strieben/cplreader_performance_suite (Erlangen) or on the CFD_Data NFS share on qnap (Klagenfurt).

7.5 EnSight

The EnSight File Format is described in Chapter 11 of the EnSight 7 User's Manual http://www-vis.lbl.gov/NERSC/Software/ensight/docs/OnlineHelp/UM-C11.pdf and here https://visualization.hpc.mil/wiki/EnSight%27s_Gold_Casefile_Format. In cplreader we make use of the EnSight file readers provided by VTK.

There exists a tool called ens_checker for checking the validity of given EnSight input files. It is a part of EnSight but may also be downloaded (for Linux) at http://aero.ist.utl.pt/~rreis/ens_checker.[32|64]. It has also been put on https://lse17.e-technik.uni-erlangen.de:2001/files/precompiled/EnSight/. A Howto for using ens_checker can be found at http://www-vis.lbl.gov/NERSC/Software/ensight/doc/OnlineHelp/HT-Read-ens_checker.pdf

7.5.1 Required File Layout

```
zyl/
|-- 20110207_Ensight_gold_binary_LES_Zylinder.encas
|-- 20110207_Ensight_gold_binary_LES_Zylinder.geo
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.scl1
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.scl2
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.scl3
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.scl4
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.scl5
|-- 20110207_Ensight_gold_binary_LES_Zylinder141221.vel
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.scl1
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.scl2
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.scl3
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.scl4
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.scl5
|-- 20110207_Ensight_gold_binary_LES_Zylinder141222.vel

...

|-- zyl.case
`-- zyl_bin.geo -> 20110207_Ensight_gold_binary_LES_Zylinder.geo
```

In this example the VTK Ensight reader could not directly read the `20110207_Ensight_gold_binary_LES_Zylinder.encas` case file due problems with the file name of the `.geo` file. Therefore a copy of it (`zyl.case`) has been created which contains the following information:

```
FORMAT
type:  ensight gold
GEOMETRY
model:  2  zyl_ascii.geo
VARIABLE
scalar per node:  1  pressure                  20110207_Ensight_gold_ASCII_LES_Zylinder****.s
scalar per node:  1  density                   20110207_Ensight_gold_ASCII_LES_Zylinder****.s
scalar per node:  1  x_velocity                20110207_Ensight_gold_ASCII_LES_Zylinder****.s
scalar per node:  1  y_velocity                20110207_Ensight_gold_ASCII_LES_Zylinder****.s
scalar per node:  1  z_velocity                20110207_Ensight_gold_ASCII_LES_Zylinder****.s
vector per node:  1  velocity                  20110207_Ensight_gold_ASCII_LES_Zylinder****.v
TIME
time set: 1 Model
number of steps: 10
filename start number:      141231
filename increment:         1
time values:  1.41231e+00  1.41232e+00  1.41233e+00  1.41234e+00  1.41235e+00  1.41236e+00  1
 1.41238e+00  1.41239e+00  1.41240e+00
time set:                   2 Model
number of steps:            1
time values:  1.41231e+00
```

The file `zyl_ascii.geo` is indeed just a symbolic link to the real mesh file `20110207_Ensight_gold_binary_LES_Zylinder`

7.5.2 Special Options

`--v[x|y|z]` Specify name of velocity field array to cplreader. If the field is a vector quantity set all three flags to the same value. If the field is broken up into scalar components specify the component names here.

`--pres` Specify the name of the pressure scalar field using this flag.

7.5.3 Logging

The `.case` gets saved to the /UserData/name.case dataset in the HDF5 file for later reference.

7.5.4 Examples

```
cplreader --type FIELDVIEW --name fluid_aachen --verbose 1 --justmesh 1 \
          --outputfields all --numsteps 1
```

7.6 FLUENT

ANSYS Fluent writes transient data into `.cas[.gz]` and `.dat[.gz]` files. If the mesh does not change then only one `.cas[.gz]` file and many `.dat[.gz]` files will be written. If the mesh changes a `.cas[.gz]` for every `.dat[.gz]` file gets written. Therefore, we support the following file layouts.

7.6.1 Required File Layout

```
NACA0012-2D/
|-- NACA0012-2D-23001.dat
|-- NACA0012-2D-23002.dat
|-- NACA0012-2D-23003.dat
`-- NACA0012-2D.cas

FAN-3D/
|-- FAN-3D-00006.cas
|-- FAN-3D-00006.dat
|-- FAN-3D-00007.cas
|-- FAN-3D-00007.dat
|-- FAN-3D-00008.cas
`-- FAN-3D-00008.dat
```

7.6.2 Special Options

7.6.3 Logging

The mapping from `.cas[.gz]` to `.dat[.gz]` for every timestep gets saved to the group /UserData/CasToDatMapping in the HDF5 file for later reference.

```
ts[0] (t=0s): NACA0012-2D.cas -> NACA0012-2D-23001.dat
ts[1] (t=1s): NACA0012-2D.cas -> NACA0012-2D-23002.dat
ts[2] (t=2s): NACA0012-2D.cas -> NACA0012-2D-23003.dat
```

7.6.4 Examples

```
cplreader --type FLUENT --justmesh 0 --numsteps -1 --verbose 1 --calcsrc 1
          --outputfields all --dim 2 --timestep 1 --name NACA0012-2D
```

7.7 FieldView

The implementation of the FieldView reader is quite basic and only properly supports reading version 3.0 unstructured files generated from STARCCM+ at the moment. If the format is needed in the future some improvements to its robustness have to be made. The documentation of the file format is not publicly available but a PDF is available on the CFS++ development server https://lse17.e-technik.uni-erlangen.de:2001/files/nightly-builds/doc/fieldview_format.pdf

7.7.1 Required File Layout

```
fluid_aachen/
|-- fluid_aachen.fv
|-- fluid_aachen.fv.fvreg
`-- fluid_aachen.fvuns -> fluid_aachen.fv
```

This tree has been generated by using the CFX export utility:

```
cfx5export -fieldview -UNS 3.0 /media/fluid_aachen/fluid_aachen.res
```

Files generated in this way, can however not be read into cplreader because of the mentioned robustness issues.

7.7.2 Special Options

7.7.3 Logging

7.7.4 Examples

7.8 STARCCM+

In order to examine .ccm files (which are indeed Boeing ADF files http://dx.doi.org/10.4271/985565) the CGNS http://www.cgns.org tools (especially adfview which is similar to hdfview) from CFSDEPS cgns may be used. To view the fields the VisIt http://www.llnl.gov/visit postprocessor may be used.

A STARCCM+ reader developed by Patrick Lorenz and Jens Grabinger is available at https://lse17.e-technik.uni-erlangen.de:2001/svn/CFS++/branches/starccm. It does however not read .ccm files but files in the Field View (cf. Sec. 7.7 and http://www.ilight.com) postprocessing format. This decision has been made due to the fact that .ccm files contain polyhedral cells, which cplreader cannot handle. The FieldView files have to be generated within STARCCM+. You may also contact Frank Schäfer mailto:frank.schaefer@de.cd-adapco.com from CD-adapco Nuremberg for further details.

The FieldView data format is also supported by the cfx5export tool of ANSYS CFX.

7.8.1 Required File Layout

7.8.2 Special Options

7.8.3 Logging

7.8.4 Examples

7.9 CGNS - CFD General Notation System

The CGNS file format can be generated by almost all commercial CFD codes. It has been found however, that few codes have support for writing CGNS files on the fly. Most often the native result files have to be converted to CGNS in an intermediate step. Another finding is that many CFD codes save the complete mesh in every CGNS time step file, which renders this format almost unusable for aeroacoustic analysis. At least for Fluent this situation should change with release 14 and ANSYS CFX presents the possibility to write all time step fields to a single CGNS file, while only saving the mesh once (or twice).

The CGNS reader is still under active development. Ideas for its future and other things to keep in mind are collected at https://lse17.e-technik.uni-erlangen.de:2001/trac/discussion/topic/14.

The reader has initially been developed to support data generated by ANSYS Fluent. Known problems and future perspectives are discussed at https://lse17.e-technik.uni-erlangen.de:2001/trac/discussion/topic/7.

7.9.1 Required File Layout

```
Ellipse4_sas_006
|-- Ellipse4_sas_006_2011.cgns
|-- Ellipse4_sas_006_2015.cgns
`-- Ellipse4_sas_006_2019.cgns
```

7.9.2 Special Options

7.9.3 Logging

7.9.4 Examples

Convert CFX transient results to CGNS:

```
cfx5export -cgns -timestep -1 Ellipse4_sas_006.res

Reading CFX5 results from <Ellipse4_sas_006.res>.
processing all domains.
processing all timesteps.
writing CGNS file to <Ellipse4_sas_006_2011.cgns>.
  writing 3D element section <Assembly 3D> containing 2607000 elements ... done
  writing 2D element section <AIR> containing 70800 faces ... done
  writing 2D element section <BOTTOM> containing 24740 faces ... done
  writing 2D element section <CYL> containing 11160 faces ... done
  writing 2D element section <INL> containing 12000 faces ... done
  writing 2D element section <OUT> containing 12000 faces ... done
  writing boundary condition <AIR> containing 70800 faces...done
  writing boundary condition <BOTTOM> containing 24740 faces...done
  writing boundary condition <CYL> containing 11160 faces...done
  writing boundary condition <INL> containing 12000 faces...done
  writing boundary condition <OUT> containing 12000 faces...done
writing CGNS file to <Ellipse4_sas_006_2015.cgns>.
  writing 3D element section <Assembly 3D> containing 2607000 elements ... done

...

done.
```

Afterwards move the generated files into a directory as described above and start cplreader:

```
cplreader --type CGNS --name Ellipse4_sas_006 --outputfields all --timestep 1e-4
```

8 Other Readers and Mesh Generators

8.1 ANSYS Mesh Converter

The ANSYS mesh converter can read either ANSYS mesh output files generated by the older mkmesh extension and convert them to *HDF5*or the the files generated by the NACS GiD/ANSYS extension. At LSE a version of cplreader is used to provide the ANSYS mkhdf5 (/home/data/programs/mkhdf5) APDL extension.

8.2 Mesh Generators

The syntax for calling cplreader is:

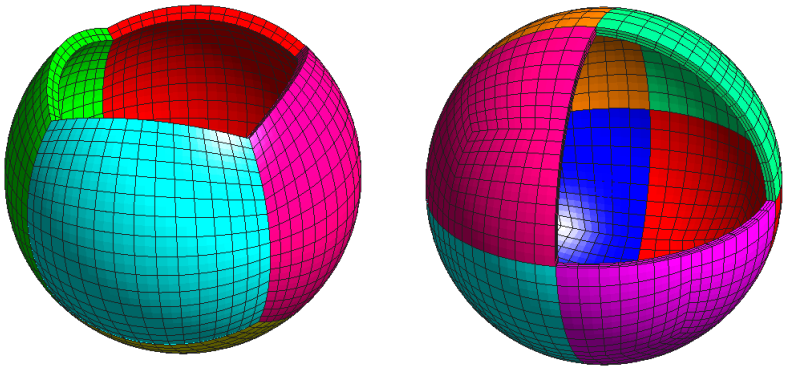
```
cplreader --name test --type GENGRIDS --xmlfile test.xml
```

8.2.1 Spherical Shells

I (Simon Triebenbacher) implemented this grid generator to try out spherical shell grids composed of all available element types (including HEXA27) for nonmatching grids. It can be only used through a settings XML file at the moment. Here is an example for an XML file for generating spherical shells.

```
<cplreaderOptions>
  <type name="GENGRIDS" subType="SphericalShellGenerator">
    <elementType name="HEXA27"/>
    <numRefinementLevels value="3"/>
    <outerRadius value="0.95"/>
    <innerRadius value="0.9"/>
    <numRadialElements value="3"/>
    <octantRegions value="off"/>
  </type>
</cplreaderOptions>
```

Figure 6 shows the influence of the 'octantRegions' parameter on the distribution of the regions inside the generated grid.



(a) Default setting (octantRegions off) (b) Using octant regions (octantRegions on)

Figure 6: Influence of the 'octantRegions' parameter.

8.2.2 Rectilinear Grids

Cplreader has been extended with a mesh generator for generating simple rectilinear meshes consisting of hexahedra elements. The geometry for the setup may only consist of a grid of boxes. The geometry is described by a number of intervals in x-, y-, and z-directions. In each interval a number of subdivisions may be specified (similar to ESIZE,,NDIV in ANSYS). Cplreader treats each of the boxes as volumes (in ANSYS terminology). The naming convention is 'volume_x_y_z' as can be seen for a simple example in Fig. 7.

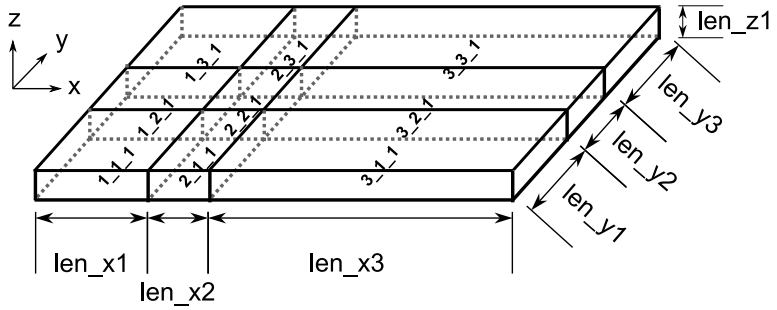


Figure 7: Naming conventions for volumes.

For each volume six surfaces are being automatically generated using the naming convention given in Fig. 8.

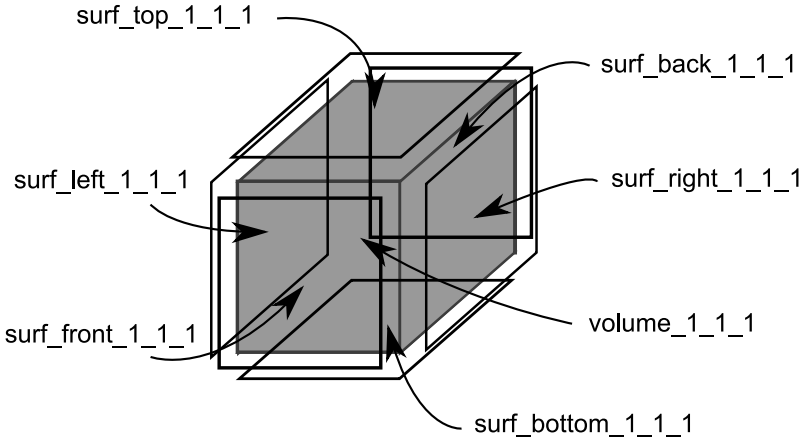


Figure 8: Naming conventions for surfaces.

For the bottom layer surface elements also line segments are create following the naming convention 'lines_[left|right|front|back]'. Due to this one can also create 2D meshes with line elements on the boundaries. Here is a sample XML file.

```
<cplreaderOptions>
  <type name="GENGRIDS">
    <elementType name="HEXA8"/>
    <samplingIntervalls>
      <!-- Define intervals by using either absolute or incremental coordinates -->
      <coordinates axis="x">
        <coord type="absolute">  0.0e-0 </coord>
        <coord type="increment"> 2.0e-1 </coord>
        <coord type="increment"> 5.0e-2 </coord>
        <coord type="increment"> 2.0e-1 </coord>
      </coordinates>
      <coordinates axis="y">
        <coord type="absolute">  0.0e-0 </coord>
        <coord type="increment"> 1.0e-1 </coord>
        <coord type="increment"> 2.0e-2 </coord>
        <coord type="increment"> 1.0e-1 </coord>
      </coordinates>
      <coordinates axis="z">
        <coord type="absolute">  0.0e-3 </coord>
        <coord type="increment"> 5e-3 </coord>
```

```
      </coordinates>
    </samplingIntervalls>
    <!-- Assign number of elements to each interval -->
    <intervallElements>
      <elements axis="x">
        <elems>2</elems>
        <elems>5</elems>
        <elems>10</elems>
      </elements>
      <elements axis="y">
        <elems>2</elems>
        <elems>7</elems>
        <elems>12</elems>
      </elements>
      <elements axis="z">
        <elems>2</elems>
      </elements>
    </intervallElements>
    <regions>
      <!-- Define regions by selecting volumes, surfaces and lines -->
      <region name="region1">
        <prelimRegion> volume_1_1_1 </prelimRegion>
        <prelimRegion> volume_2_1_1 </prelimRegion>
        <prelimRegion> volume_3_1_1 </prelimRegion>

        <prelimRegion> volume_1_2_1 </prelimRegion>
        <prelimRegion> volume_3_2_1 </prelimRegion>

        <prelimRegion> volume_1_3_1 </prelimRegion>
        <prelimRegion> volume_2_3_1 </prelimRegion>
        <prelimRegion> volume_3_3_1 </prelimRegion>
      </region>

      <region name="region2">
        <prelimRegion> volume_2_2_1 </prelimRegion>
      </region>

      <region name="surf_region2_bottom">
        <prelimRegion> surf_bottom_2_2_1 </prelimRegion>
      </region>

      <region name="left_lines">
        <prelimRegion> lines_left_1_1_1 </prelimRegion>
        <prelimRegion> lines_left_1_2_1 </prelimRegion>
        <prelimRegion> lines_left_1_3_1 </prelimRegion>
      </region>
    </regions>
  </type>
</cplreaderOptions>
```

Figure 9 shows the grid generated from the above XML file.

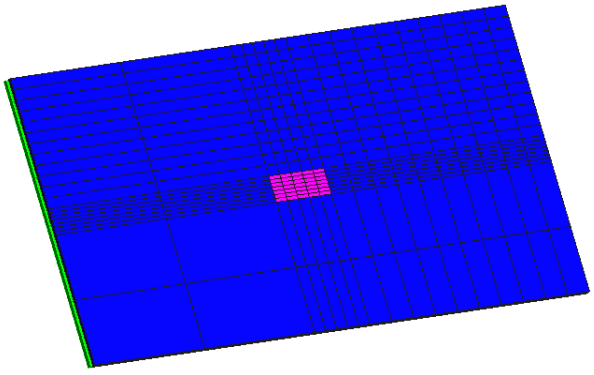


Figure 9: Rectilinear grid generated from sample XML file.

8.2.3 Reference Elements

This generator just outputs a single reference element of specified type using the coordinates and node numbering as used internally within CFS++. The following gives an example of generating a wedge element.

```
<cplreaderOptions>
  <type name="GENGRIDS" subType="ReferenceElementGenerator">
    <elementType name="WEDGE15"/>
  </type>
</cplreaderOptions>
```

Figure 10 shows the available element types file.

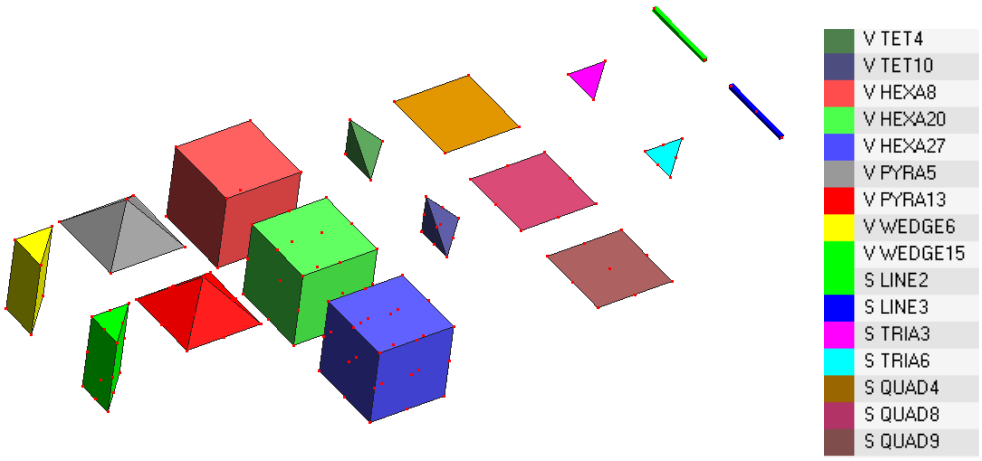


Figure 10: Array of reference elements from CFS++.

8.2.4 Defining Analytical Fields using MuParser

```
<cplreaderOptions>
<type name="GENGRIDS">
<elementType name="HEXA8"/>
<samplingIntervalls>
<!-- Define intervals by using either absolute or incremental coordinates -->
<coordinates axis="x">
<coord type="absolute"> -1.0e-0 </coord>
<coord type="increment"> 2.0e-0 </coord>
</coordinates>
<coordinates axis="y">
<coord type="absolute"> -1.0e-0 </coord>
<coord type="increment"> 2.0e-0 </coord>
</coordinates>
<coordinates axis="z">
<coord type="absolute"> 0.0e-3 </coord>
<coord type="increment"> 5e-3 </coord>
</coordinates>
</samplingIntervalls>
<!-- Assign number of elements to each interval -->
<intervallElements>
<elements axis="x">
<elems>41</elems>
</elements>
<elements axis="y">
<elems>41</elems>
</elements>
<elements axis="z">
<elems>1</elems>
</elements>
</intervallElements>
<regions>
<!-- Define regions by selecting volumes, surfaces and lines -->
<region name="fluid">
<prelimRegion> surf_bottom_1_1_1 </prelimRegion>
</region>
</regions>

    <fields>
        <field name="acouRhsLoad">
            <component axis="x">
                x*if(t lt .005,t/.005,1)
            </component>
        </field>

        <field name="fluidMechVelocity">
            <component axis="x">
                -y/sqrt(x^2+y^2)*if(t lt .005,t/.005,1)
                * if( ((x - 0.5)^2 + (y - 0.5)^2) lt 0.1, 0, 1)
                * if( ((x + 0.5)^2 + (y + 0.5)^2) lt 0.1, 0, 1)
            </component>
            <component axis="y">
                x/sqrt(x^2+y^2)*if(t lt .005,t/.005,1)
                * if( ((x - 0.5)^2 + (y - 0.5)^2) lt 0.1, 0, 1)
                * if( ((x + 0.5)^2 + (y + 0.5)^2) lt 0.1, 0, 1)
            </component>
            <component axis="z">
                0
            </component>

        </field>

    </fields>

</type>
</cplreaderOptions>
```

Figure 11 shows the resulting velocity field.

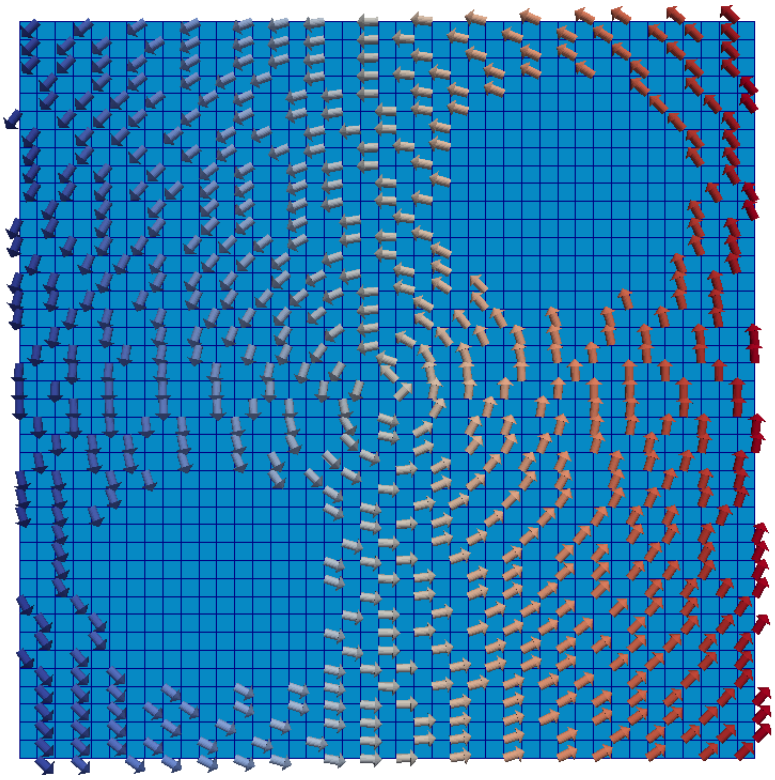


Figure 11: Velocity field generated by cplreader.

The version of muParser which is used in cplreader does not support complex arithmetic. This means that transient phenomena modelled by complex expressions must be translated to the corresponding sine or cosine expressions by hand. For details concerning complex arithmetic support in muParser cf. https://lse17.e-technik.uni-erlangen.de:2001/trac/discussion/topic/10.

9 Setting up the Interpolation Calculation in CFS++

In order to interpolate the acoustic source terms from the CFD mesh, which cplreader has written out, to the acoustic mesh of the source region a CFS++ simulation has to be set up. One needs to have a XML file, a material file and the mesh files for the acoustic mesh and the CFD mesh with the acoustic sources. A typical XML file for doing the interpolation looks like this:

```
<?xml version="1.0"?>

<cfsSimulation xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://www.cfs++.org
 /opt/pckg/cfs_nightly/trunk_gcc/share/xml/CFS-Simulation/CFS.xsd"
 xmlns="http://www.cfs++.org">

  <fileFormats>
    <input>
      <!-- Mesh of acoustic source region      -->
      <hdf5 fileName="2d_fluidregion.h5"
            linearizeEntities="all"
            coordSysId="myCoordSys"/>

      <!-- Mesh from CFX plus acoustic source terms generated by cplreader -->
      <hdf5 fileName="test.h5" id="ipolSrc" gridId="interpolGrid"/>
    </input>

    <output>
       <hdf5 id="hdf5"/>
    </output>

    <materialData file="mat.xml" format="xml"/>
  </fileFormats>

  <domain geometryType="3d">
    <regionList>
      <region name="fluid" material="air"/>
    </regionList>

    <coordSysList>
      <!--   Use   this   coordinate   system   to   remap   the   geometry   in
           2d_fluidregion.h5 before interpolation -->
      <trivialCartesian id="myCoordSys">
        <origin x="-5e-4"/> <axisMap x="x" y="y" z="z"/>
      </trivialCartesian>
    </coordSysList>
  </domain>

  <sequenceStep>
    <analysis>
      <transient>
         <numSteps> 9 </numSteps>
        <deltaT> 3.125e-4 </deltaT>
      </transient>
    </analysis>

    <pdeList>
      <!-- name of pde -->
      <acoustic formulation="acouPotential">

        <regionList>
          <region name="fluid"/>
        </regionList>

        <bcsAndLoads>
          <!--  This section is  important for  the interpolation!!!   It says
                that the  source terms  are located in  region "fluid"  on the
                acoustic  grid  and  that  they should  be  interpolated  from
                regions  "partition1-3" from the  source grid  with srcInputId
                "ipolSrc".  The  tolerance parameters control  some aspects of
                the conservative interpolation. Once the interpolation weights
```

```
                for a  pair of grids have been  computed it is a  good idea to
                cache  them  in  a  file  and just  read  them  in  subsequent
                simulations.  -->
            <rhsValues region="fluid">
              <factor>1</factor>
              <interpolation>
                <srcRegions names="partition1 partition2 partition3"
                            inputId="ipolSrc"/>

                <!-- There are  global and local tolerances for  finding the CFD
                     nodes inside  the acoustic mesh.  If some  nodes cannot get
                     mapped using the start  tolerence the next higher tolerance
                     will be tried up to the end tolerance in inc steps. -->
                <tolerances>
                    <global start="1e-6" inc="1e-7" end="4e-6"/>
                    <local  start="1e-4" inc="1e-5" end="5e-4"/>
                </tolerances>

                <!-- Display warnings about nodes which could not be mapped. -->
                <nodeWarnings display="verbose"/>

                <!-- For  large CFD meshes  it is a  good idea to use  a restart
                file   which   contains   all  previously   found   conservative
                interpolation weights. If  not all CFD nodes could  be mapped in
                the first run the restartFileMode can be set to rw, which causes
                CFS++  not to  recompute  the weights  which  have already  been
                found. -->
                <restartFileMode>w</restartFileMode>
              </interpolation>
            </rhsValues>
          </bcsAndLoads>

          <storeResults>
            <!--  Since  we  want  to   do  a  transient  simulation  using  the
                  interpolated right  hand side values  we save them  to another
                  HDF5 file.  ( results_hdf5/output_of_interpolation.h5 ) -->
            <nodeResult type="acouRhsLoad">
              <regionList saveBegin="1" saveInc="1" saveEnd="9">
                <region name="fluid"/>
              </regionList>
            </nodeResult>
          </storeResults>
        </acoustic>
      </pdeList>

      <linearSystems>
        <!-- The  settings for  the linear systems  are not important  since the
             acoustic PDE does not have to be solved while interpolating -->
        <system name="acoustic">
          <setup idbcHandling="penalty"/>
          <matrix entry="double"
                  storage="sparseNonSym"
                  reordering="noReordering"/>
          <!--exportLinSys baseName="out"/-->
          <solver type="gmres"
                  precond="Id"/>
        </system>
      </linearSystems>
    </sequenceStep>
</cfsSimulation>
```

Figure 12 shows the meaning of the tolerance parameters in the interpolation section of the XML file. In order to find the correct destination element/cell for a given node of the CFD mesh we use the Intersecting Sequences of dD Iso-oriented Boxes algorithm from the Computational Geometry Algorithms Library (CGAL) [1]. This algorithm takes two sets of bounding boxes, does some spatial sorting on them and calls a callback function for each pair of intersecting boxes from the two sets. We use this algorithm by generating axis aligned bound boxes around the cells of the acoustic mesh as well as around the nodes of the CFD mesh. The bounding boxes around the acoustic cells are generated by find the minimal and maximal coordinates of the nodes of the acoustic cell. The bounding boxes for the CFD nodes are generated by using the global tolerance parameter from the interpolation in

a way shown in Fig. 12. If such a pair of bounding boxes intersects the callback function will then transform the location of the CFD node into the reference coordinate system of the acoustic cell and use the local tolerance parameter to specify a region around the reference cell. If the CFD node is within this region the algorithm will then use the current acoustic cell as destination cell. The value of the CFD node will just be distributed among the nodes of this acoustic cell.

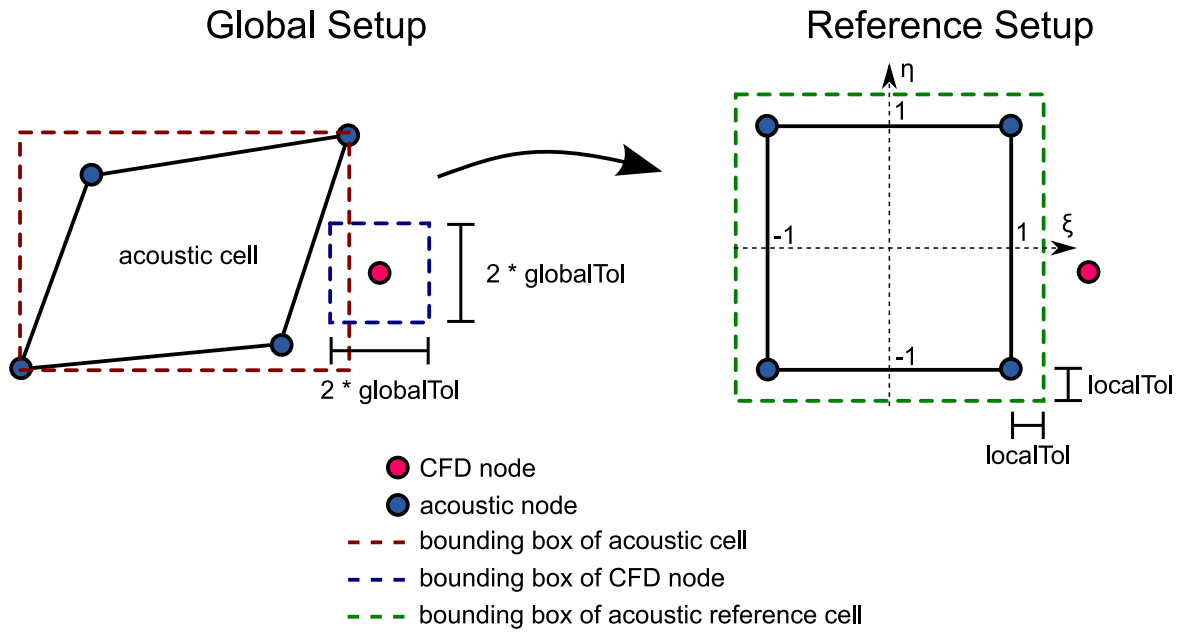


Figure 12: Meanings of the tolerance parameters for the interpolation step.

The algorithm would select the bounding box pair of the CFD node and the acoustic cell and do a callback for the example shwon in Fig. 12. It would however reject to map the CFD node to the acoustic cell since the node is outside the local tolerance. A good guidline for choosing the tolerances is to use less than half of the minimal edge length in the acoustic grid as maximal global tolerance so that only nodes really close to the acoustic cells get selected by the intersecting bounding boxes algorithm. Specifying a too large value here will cause lots of unsuccessful bounding box intersection operations. It is however sometimes necessary to go up with the global tolerance to be able to map on curved boundaries. One should choose as small local tolerances as possible since the the basis functions for weighting the source value tend to be negative outside an elements domain. Another cause for errors is the fact that the CFD node can get projected to the wrong destination cell if this tolerance is to large. The cell neighboring the current cell my be the correct one, but the node already got mapped. This is why one should start the interpolation with very tight global and local tolerances and iteratively add CFD nodes which need larger tolerances. This may either be achieved by using the start, end and inc attributes in the XML file or by making use of the restart file or by combining both possibilities.

9.1 Important Guidelines

Allways check if the sum of the source terms on the source and destination meshes are equal. The only case when the sums should not be equal is when you interpolate from a geometrically larger source domain into a smaller destination domain. If the sum over the the source domain cannot be computed easily (e.g. when interpolating just one plane of a 3D dataset to 2D) try to evalute the sum for a number of different destination meshes. In the simplest case just use one destination element covering the whole source region. The evaluation of the sum may be performed by reading in the datasets

`/Results/Mesh/MultiStep_1/Step_X/acouRhsLoad/REGIONNAME/Nodes/Real`

on the source and destination HDF5 files and summing them up. This may be done by the following Matlab code

```
fileName = 'file.h5'
dsName = '/Results/Mesh/MultiStep_1/Step_X/acouRhsLoad/REGIONNAME/Nodes/Real'

vec = hdf5read(fileName, dsName);

sum = 0
for i=1:length(vec)
  sum = sum+vec(i);
end
```

Never use the text output of h5dump[6] or the CSV output of ParaView to do comparisons since both programs cut down the digits of the numbers. One can use h5dump to extract the binary data however by issuing

```
DS=/Results/Mesh/MultiStep_1/Step_X/acouRhsLoad/REGIONNAME/Nodes/Real
h5dump -o out.bin -b MEMORY -d $DS file.h5
```

The resulting binary data may be read by Octave using the following script

```
f = fopen('out.bin', "r");
% Specify here the length of the vector from the HDF5 file.
sizeVec = 10;
vec = fread (f, sizeVec, "double", 0);

function s = sumUp(m,n)
s = 0;
for i=1:n
  s = s + m(i);
end
endfunction

sumUp(vec,sizeVec)
```

9.2 Getting into the Code

The code for the calculation of the conservative interpolation is distributed among the following files:

```
source/Forms/acouRHSLinForm.cc
source/Domain/grid.hh
source/Domain/grid.cc
```

The following methods are of interest:

- `AcouRHSLinForm::CalcElemVector (acouRHSLinForm.cc)` Calculates exactly one entry of the acoustic right hand side vector.
- `Grid::ComputeConservativeInterpolationWeights (grid.cc)` Computes bounding boxes for source nodes and destination elements for one source region and calls the bounding box intersection algorithm.
- `Grid::ConsInterpReportFunctor::operator() (grid.hh)` Is a callback for the bounding box algorithm and checks if a source node is indeed inside a destination element. If this is the case it calculates the interpolation weights by evaluating the element basis functions at the local coordinates of the source node.

[6] http://www.speedup.ch/workshops/w37_2008/HDF5-Tutorial-PDF/HDF5-Tools.pdf

10 Rebuilding the Documentation

Switch on `CPLREADER_REBUILD_DOCU` in CMake GUI. Make sure you have a recent TikZ/PGF distribution and Pygments (`http://pygments.org` → `sudo easy_install Pygments`) installed. Edit the following files:

```
source/cplreader/Documentation/latex/CMakeLists.txt
source/cplreader/Documentation/latex/cplreader_docu.tex
source/cplreader/Documentation/latex/Title.tex
source/cplreader/cmake_options.cmake
```

To add additional sources edit the CMakeLists.txt and add them to the list of CPLREADER_DOCU_SOURCES. After doing so make sure to do a CMake run so that the sources get copied to the build directory. This step is also necessary after copying new pictures to the pics subdirectory. Then you may rebuild CFS++ and cplreader. The resulting PDF documentation file for cplreader will be copied back to the source directory automatically. Therefore you just need to commit the changes to SVN.

11 Copyright

12 See Also

Homepage of the LSE: http://www.lse.uni-erlangen.de

Homepage of the Chair for Applied Mechatronics:
http://www.uni-klu.ac.at/tewi/ict/sst/am/index.html

Homepage of the Simetris GmbH: http://www.simetris.de

Hierarchical Data Format page: http://hdf.ncsa.uiuc.edu/HDF5

OpenFOAM: http://www.opencfd.co.uk/openfoam

ANSYS CFX: http://www.ansys.com/Products/cfx.asp

Computational Aeroacoustics in Erlangen: http://www.lstm.uni-erlangen.de/aeroakustik/

Computational Fluid Dynamics at LSTM Erlangen
http://www.lstm.uni-erlangen.de/Ber/BerT1/index-en.htm

Chair of Fluid Mechanics and Fluid Machines, University Freiberg
http://www.imfd.tu-freiberg.de/fluid/index.en.html

Homepage of István Vaik (user of cplreader for edge tone simulations)
http://www.vizgep.bme.hu/okto/vi/index.html

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